

Weak Nodes and Weak Links Are Not Graph-Centrality Objects in IBR Grids

Jorge L. Mayorga Taborda
Universidad de los Andes, Bogota, Colombia
Email: author@example.com

Abstract—Weak nodes and weak links in inverter-based-resource (IBR) grids are often discussed as if they were graph objects: highly central buses, heavily loaded branches, low short-circuit-strength locations, or large entries of a Laplacian eigenvector. This paper argues that this is the wrong object for damping-margin weakness once inverter and controller states participate in the critical mode. The reduced object is a controller-aware nonlinear eigenvalue problem (NEP), obtained by Schur-complement condensation of controller states. In this representation, a line or node is weak only relative to an action, a pole family, and the frequency-dependent controller self-energy. We formalize this statement with a local NEP sensitivity certificate and show that a graph-only centrality score cannot determine the sign of a damping-margin change when condensed controller poles are present. The same framework separates ordinary damping-ratio geometry from a stricter *control-resonant Braess* mechanism: strengthening a line may reduce damping because it moves a network-family mode toward a PLL-dominated controller pole. On IEEE 39-bus ANDES cases, a Beyn contour solver for the Schur NEP reproduces the full ANDES critical poles to numerical precision, including a 13.742 Hz PLL-family Mix60 mode missed by scalar reductions. A registered line audit over 460 perturbations finds 40 strict resonant records; the clearest case is Line 20–34 in Mix60/no-PSS, where the self-energy term predicts the network-mode damping decrease with 0.2% relative error. The contribution is not a new graph centrality, but a dynamic certificate explaining when graph centrality is the wrong language.

Index Terms—Inverter-based resources, damping margin, weak nodes, weak links, Braess paradox, nonlinear eigenvalue problem, Schur complement, phase-locked loop.

I. INTRODUCTION

Power-grid planning often uses graph language. A weak bus is associated with electrical distance, short-circuit strength, modal participation, or centrality; a weak line is associated with loading, cut structure, or Laplacian sensitivity. These ideas remain useful for static network questions, but they can become misleading for damping-margin questions in inverter-dominated grids. The reason is simple: in detailed IBR models, the damping observed in the least-damped pole may be supplied by PLL, exciter, stabilizer, governor, grid-following, or grid-forming controller states rather than by a static nodal damping coefficient.

This paper develops a narrow alternative. A weak element is not defined by where it sits in the graph. It is defined by how a physically admissible action moves the damping-margin pole of the controller-aware reduced eigenproblem. To keep the object transparent, we derive the certificate from the

electrical power-flow equations before introducing the Schur reduction. The result is a local certificate:

$$\begin{aligned} \text{weak element} = & \text{negative pole-margin sensitivity} \\ & + \text{control-resonance exposure.} \end{aligned}$$

This viewpoint is especially important for Braess-like claims. If a line reinforcement raises oscillation frequency while the decay rate does not increase enough, the damping ratio can fall. That is a geometric damping-ratio effect, not necessarily a network paradox. A stricter and more IBR-specific mechanism appears when reinforcement moves a network-family mode toward a nearby controller pole, so the frequency-dependent self-energy of the controller dominates the damping loss. We call this a control-resonant Braess mechanism.

The paper makes three contributions.

- 1) It defines controller-aware weak-node and weak-link certificates from Schur-NEP pole sensitivities rather than graph centrality.
- 2) It proves a non-identifiability result: graph-only centrality cannot determine weak elements when the controller self-energy is active, because the graph can be fixed while the sign of the damping sensitivity changes through condensed controller poles.
- 3) It gives a minimal three-bus witness and validates the mechanism on IEEE 39-bus ANDES artifacts: the Schur NEP reproduces full ANDES critical poles, and a line audit finds strict resonant reinforcement reversals under a PLL-dominated self-energy term.

The scope is deliberately limited. We do not claim that all line-reinforcement reversals are resonant, that graph centrality is useless for static planning, or that the proposed certificate is already a full planning optimizer. The result is a mechanism paper: it identifies the dynamic object that graph centrality misses. Figs. 1–2 also include the associated inertia warning from the project gates: adding inertia can reduce damping in GENROU/reduced-model tests, but GFM virtual-inertia assignment over the controller-aware bridge remains a separate validation task.

II. FROM ELECTRICAL MODEL TO DAMPING CERTIFICATE

This section derives the object used in the paper from the standard small-signal power-system equations. The purpose is not to introduce a new network model, but to show exactly where the usual graph object stops being sufficient.

Failure mode: a stronger static graph can have a smaller damping margin

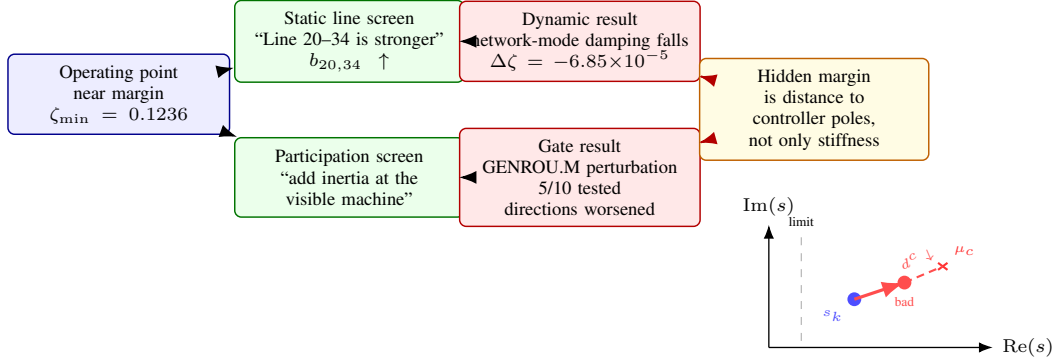
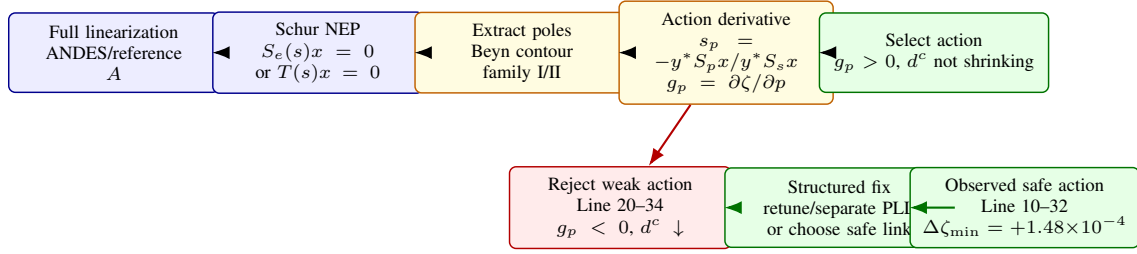


Fig. 1. Why Fig. 2 is needed. Static planning signals can point in the wrong direction for damping: a line reinforcement may increase synchronizing stiffness while reducing the damping ratio, and an inertia perturbation can worsen damping in the validated GENROU gate. The common failure is that neither static graph strength nor modal visibility measures distance to condensed controller poles.

Controller-aware workflow: do not rank the graph; move the pole safely



Engineering reading: if a proposed reinforcement or inertia change moves the tracked pole rightward, reduces ζ , or decreases the distance to a PLL/control pole, the action is not a robustness upgrade even if the static graph looks stronger.

Fig. 2. Controller-aware robustness workflow. The certificate turns weak-node and weak-link analysis into a typed engineering statement: the element is weak for a specified mode, action, and controller configuration.

A. Electrical Network and Linearized Power Flow

Step 1: represent each bus by a phasor. At a bus i , the balanced positive-sequence voltage is represented by the phasor

$$V_i = |V_i|e^{j\theta_i}, \quad (1)$$

where $|V_i|$ is the voltage magnitude and θ_i is the phase angle with respect to the synchronous reference. The time waveform is recovered as $v_i(t) = \sqrt{2} \text{Re}\{V_i e^{j\omega_0 t}\}$. This phasor representation is important because electromechanical oscillations and PLL dynamics are primarily angle/frequency phenomena: the small variables are not the absolute voltages, but the angle deviations $\Delta\theta_i$ and frequency deviations $\Delta\omega_i = \Delta\dot{\theta}_i$.

Step 2: write nodal current and complex power. Let $Y = G + jB$ be the network admittance matrix. The complex current injection is $I = YV$, i.e.,

$$I_i = \sum_{j=1}^n Y_{ij} V_j. \quad (2)$$

The complex power injection at bus i is

$$S_i = P_i + jQ_i = V_i I_i^*. \quad (3)$$

Expanding the active-power component gives

$$P_i = \sum_{j=1}^n |V_i| |V_j| [G_{ij} \cos(\theta_i - \theta_j) + B_{ij} \sin(\theta_i - \theta_j)]. \quad (4)$$

This is the nonlinear AC network equation. The dynamic model below uses its Jacobian at the operating point.

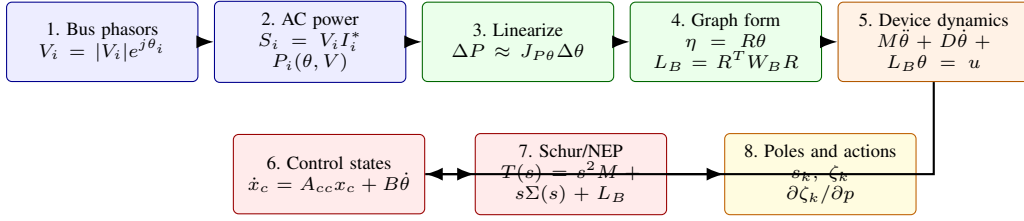
Step 3: linearize active power with respect to angles. For high-voltage transmission lines, conductance is small compared with susceptance, so the active-power sensitivity to angle is dominated by the susceptive term. Linearizing (4) around an operating point θ^0 gives, for a line (i, j) ,

$$\Delta P_{ij} = b_{ij}(\Delta\theta_i - \Delta\theta_j), \quad b_{ij} = |V_i| |V_j| B_{ij} \cos(\theta_i^0 - \theta_j^0). \quad (5)$$

Thus every line measures an angle difference. This is the physical reason a graph Laplacian appears. Equivalently, the active-power Jacobian has the local entries

$$\left. \frac{\partial P_i}{\partial \theta_j} \right|_{\theta^0} = \begin{cases} -b_{ij}, & i \neq j, \\ \sum_{\ell \neq i} b_{i\ell}, & i = j, \end{cases} \quad (6)$$

up to the chosen sign convention for injection versus imbalance.



Tutorial roadmap. The first four boxes are the familiar electrical graph construction. The last four boxes are the dynamic extension needed for IBR damping margins. The paper's weak-node and weak-link certificates live in box 8, not in graph centrality alone.

Fig. 3. Step-by-step route from phasors to the controller-aware damping certificate.

Step 4: stack line differences with an incidence matrix.

To make the graph construction explicit, orient the m lines arbitrarily and let $R \in \mathbb{R}^{m \times n}$ be the incidence matrix,

$$R_{\ell i} = 1, \quad R_{\ell j} = -1 \quad \text{for line } \ell = (i, j). \quad (7)$$

Then the vector of line angle drops is

$$\eta = R\theta, \quad (8)$$

and the vector of linearized line-power flows is

$$f = W_B R\theta, \quad W_B = \text{diag}(b_\ell). \quad (9)$$

Nodal active-power imbalance is the signed sum of incident line flows:

$$\Delta P_e = R^T f = R^T W_B R \Delta \theta = L_B \Delta \theta. \quad (10)$$

Therefore

$$L_B = R^T W_B R, \quad (11)$$

or, entrywise,

$$(L_B)_{ij} = \begin{cases} \sum_{\ell \neq i} b_{\ell i}, & i = j, \\ -b_{ij}, & i \neq j \text{ and line } (i, j) \text{ exists,} \\ 0, & \text{otherwise.} \end{cases} \quad (12)$$

The graph is therefore not an abstract analogy: it is the matrix form of linearized active-power balance. The nodes are bus phasor angles, the edges are line susceptance-weighted angle differences, and the Laplacian maps angle deviations into electrical power imbalance.

This is also the point where a graph-only analysis usually enters. A line reinforcement increases one weight b_ℓ in W_B , hence increases a positive semidefinite rank-one term

$$\Delta L_B = \Delta b_\ell r_\ell^T r_\ell, \quad (13)$$

where r_ℓ is the incidence row of the line. If only (10) is inspected, a larger b_ℓ looks like a stronger synchronizing network. The rest of the paper shows why this conclusion can be incomplete for damping.

B. Swing and Grid-Forming Interface Dynamics

Step 5: connect electrical imbalance to device motion. For a synchronous generator, or a grid-forming inverter represented by a virtual swing equation, the retained electromechanical variables satisfy

$$\dot{\theta}_i = \omega_i, \quad (14)$$

$$M_i \dot{\omega}_i = -(L_B \theta)_i - D_i \omega_i + u_i, \quad (15)$$

where M_i is physical or virtual inertia, D_i is local damping or droop, and u_i collects controller and auxiliary feedback channels not yet modeled. This is the small-signal power-balance law: inertial power equals mechanical/control injection minus electrical imbalance minus damping. In vector form,

$$M \ddot{\theta} + D \dot{\theta} + L_B \theta = u. \quad (16)$$

If $u = 0$, this is a quadratic dynamical system. Introduce $\omega = \dot{\theta}$ and $x_{\text{em}} = [\theta^T, \omega^T]^T$. The corresponding first-order state matrix is

$$\frac{d}{dt} \begin{bmatrix} \theta \\ \omega \end{bmatrix} = A_{\text{em}} \begin{bmatrix} \theta \\ \omega \end{bmatrix}, \quad (17)$$

$$A_{\text{em}} = \begin{bmatrix} 0 & I \\ -M^{-1} L_B & -M^{-1} D \end{bmatrix}. \quad (18)$$

The poles of the retained electromechanical model are the eigenvalues of A_{em} , equivalently the roots of the quadratic eigenvalue problem

$$\det(s^2 M + s D + L_B) = 0. \quad (19)$$

If D is proportional to M , the modes decouple in the eigenvectors of the mass-normalized Laplacian $M^{-1/2} L_B M^{-1/2}$. With $q_k^* M q_\ell = \delta_{k\ell}$ and $q_k^* L_B q_k = \nu_k$, the modal damping ratio is

$$\zeta_k = \frac{\delta_k}{2\sqrt{\nu_k}}, \quad \delta_k = q_k^* D q_k. \quad (20)$$

This scalar formula is useful, but it already shows a warning: increasing stiffness ν_k does not automatically increase damping ratio unless the modal damping δ_k increases enough.

C. Controller States and the Hidden Channel

Step 6: add the controller states that the graph does not see. IBR models add states that are not present in (16): PLL filters, current-control states, outer-loop states, virtual-oscillator states, power controllers, exciters, stabilizers, governors, and measurement filters. Let x_c denote those states and suppose that, after linearization, they interact with the retained network variables as

$$M\ddot{\theta} + D_0\dot{\theta} + L_B\theta + Kx_c = 0, \quad (21)$$

$$\dot{x}_c = A_{cc}x_c + B\dot{\theta}. \quad (22)$$

Step 7: eliminate the controller states in the Laplace domain. Taking Laplace transforms of (21)–(22) gives

$$(s^2M + sD_0 + L_B)\theta(s) + Kx_c(s) = 0, \quad (23)$$

$$(sI - A_{cc})x_c(s) = sB\theta(s). \quad (24)$$

For $s \notin \text{spec}(A_{cc})$,

$$x_c(s) = s(sI - A_{cc})^{-1}B\theta(s). \quad (25)$$

Substituting (25) into (23) gives

$$[s^2M + s(D_0 + K(sI - A_{cc})^{-1}B) + L_B]\theta(s) = 0. \quad (26)$$

The bracketed matrix is a nonlinear eigenvalue problem (NEP). The term

$$\Sigma(s) = D_0 + K(sI - A_{cc})^{-1}B \quad (27)$$

is the controller self-energy. It is frequency dependent and has poles at $\text{spec}(A_{cc})$. Therefore the margin is not determined only by the graph stiffness L_B or by a constant damping matrix. A line, inertia, or controller change is dangerous when it moves a relevant zero of (26) toward a low-damped controller pole or rotates the mode into a poorly damped controller channel.

D. Why Poles, Not Only Eigenvectors, Define Stability Margin

Step 8: define stability from poles. For any linearized model, a mode has the time dependence $e^{st}v$. If $s = \sigma + j\omega$, then σ gives the exponential decay rate and $|\omega|/(2\pi)$ gives the oscillation frequency. The mode is stable only when $\sigma < 0$. Its damping ratio is

$$\zeta(s) = -\frac{\sigma}{\sqrt{\sigma^2 + \omega^2}}. \quad (28)$$

The damping margin is the minimum $\zeta(s)$ over the relevant oscillatory poles. This is why a weak element cannot be defined by an eigenvector entry alone. An eigenvector says where a mode is visible; the pole says whether the mode decays, how fast it decays, and how close it is to a controller resonance.

In the scalar graph model, strengthening a line changes only L_B and therefore changes modal stiffness. In the controller-aware model, the same action changes the zero of (26) through both L_B and the resolvent $(sI - A_{cc})^{-1}$. When s is close to a controller pole $\mu \in \text{spec}(A_{cc})$, small physical changes can produce large pole motion because the denominator $s - \mu$ is small. This is the hidden-margin mechanism: the operating

point can look acceptable in a graph or frequency-only screen while the damping pole is already close to a control resonance.

The remainder of the paper uses the same object in a more general state-space form and then turns it into weak-link and weak-node certificates.

III. CONTROLLER-AWARE REDUCTION

Let the linearized full small-signal model be

$$\dot{x} = \begin{bmatrix} A_{ee} & A_{ec} \\ A_{ce} & A_{cc} \end{bmatrix} \begin{bmatrix} x_e \\ x_c \end{bmatrix}, \quad (29)$$

where x_e denotes retained network/interface states, and x_c denotes condensed controller and auxiliary states. In the ANDES experiments, x_c includes PLL, inverter-control, exciter, PSS, governor, and frequency-measurement states. For $s \notin \text{spec}(A_{cc})$, eliminating x_c gives the Schur NEP

$$S_e(s)x_e = [sI - A_{ee} - A_{ec}(sI - A_{cc})^{-1}A_{ce}]x_e = 0. \quad (30)$$

Equivalently, after reducing the retained angle/frequency coordinates to second-order form, the same physics can be written as

$$T(s)x = [s^2M + s\Sigma(s) + \tilde{L}]x = 0, \quad (31)$$

$$\Sigma(s) = D_0 + C(sI - A_{cc})^{-1}B.$$

The matrix $\Sigma(s)$ is a rational controller self-energy. A constant damping matrix is recovered only under a fast-control limit and away from singularities of $sI - A_{cc}$. Near a controller pole, the frequency dependence is not a correction; it is the mechanism.

The sensitivity formula used in the rest of the paper is just implicit differentiation of the NEP. Write the reduced equation as

$$S_e(s, p)x(s, p) = 0, \quad (32)$$

where p is a physical action parameter: a line susceptance, an inertia value, a damping gain, or a controller tuning. Let s_k be a simple zero with right null vector x_k and left null vector y_k , so that

$$S_e(s_k, p)x_k = 0, \quad y_k^*S_e(s_k, p) = 0. \quad (33)$$

Differentiate (32) with respect to p :

$$\left(S_s(s_k) \frac{\partial s_k}{\partial p} + S_p(s_k) \right) x_k + S_e(s_k, p) \frac{\partial x_k}{\partial p} = 0. \quad (34)$$

Left-multiplying by y_k^* removes the unknown vector derivative because $y_k^*S_e(s_k, p) = 0$. Therefore the pole derivative is

$$\frac{\partial s_k}{\partial p} = -\frac{y_k^*S_p(s_k)x_k}{y_k^*S_s(s_k)x_k}, \quad (35)$$

where

$$S_s(s) = I + A_{ec}(sI - A_{cc})^{-2}A_{ce}. \quad (36)$$

To see the sign in (36), differentiate the Schur term explicitly:

$$\frac{\partial}{\partial s} [-A_{ec}(sI - A_{cc})^{-1}A_{ce}] = A_{ec}(sI - A_{cc})^{-2}A_{ce}. \quad (37)$$

Thus the denominator in (35) is not a numerical artifact. It is the slope of the reduced eigenproblem at the pole. When that

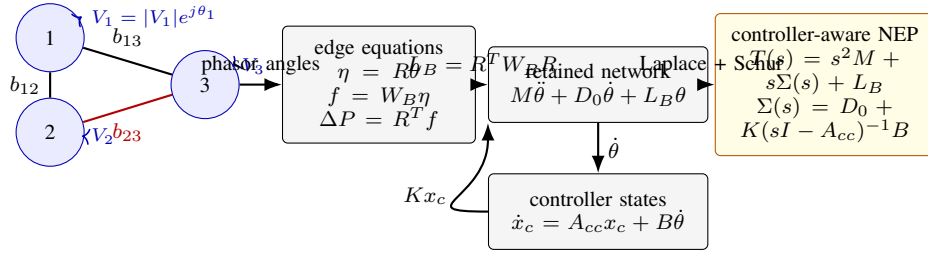


Fig. 4. Electrical origin of the controller-aware object. The graph Laplacian L_B comes from linearized line flows. IBR and legacy control states add the frequency-dependent self-energy $\Sigma(s)$; weak elements are actions that move zeros of this combined object, not static graph nodes or edges alone.

slope is dominated by controller dynamics, the same physical perturbation can produce a much larger pole motion than a static graph model predicts. This derivation is the reason weak elements are action-conditioned: the same mode s_k can move in different directions for different $S_p(s_k)$.

For the second-order form (31), the same calculation gives

$$T_s(s) = 2sM + \Sigma(s) + s\Sigma'(s), \quad \Sigma'(s) = -C(sI - A_{cc})^{-2}B. \quad (38)$$

The action derivative is, schematically,

$$T_p(s) = s^2M_p + s\Sigma_p(s) + \tilde{L}_p. \quad (39)$$

For example, in the graph-only part of the model, strengthening line $e = (i, j)$ with incidence row r_e gives

$$\frac{\partial L_B}{\partial b_e} = r_e^T r_e. \quad (40)$$

In a full controller-aware linearization, however, the same physical action may also perturb the Schur blocks and hence the self-energy term $\Sigma_p(s)$. The term with $(sI - A_{cc})^{-2}$ is the controller-resonance signature: it grows when a retained-network zero approaches a condensed-control pole.

The damping ratio of a pole $s = \alpha + j\omega$ is

$$\zeta(s) = -\frac{\text{Re}(s)}{|s|}. \quad (41)$$

To obtain the damping-ratio derivative, let $s_p = \partial s / \partial p$ and $r = |s|$. Since $\partial r / \partial p = \text{Re}(\bar{s}s_p) / r$,

$$\frac{\partial}{\partial p} \left(-\frac{\text{Re}(s)}{r} \right) = -\frac{\text{Re}(s_p)}{r} + \frac{\text{Re}(s)}{r^2} \frac{\partial r}{\partial p}. \quad (42)$$

Substituting the expression for $\partial r / \partial p$ gives

$$\frac{\partial \zeta}{\partial p} = -\frac{\text{Re}(s_p)}{|s|} + \frac{\text{Re}(s) \text{Re}(\bar{s}s_p)}{|s|^3}. \quad (43)$$

Equations (35)–(43) are the local object used below to define weak nodes and weak links.

How an engineer reads the final equations. Equation (35) answers the question that a planning study actually asks: if I change this line, this inertia, or this controller gain, which way does the critical pole move? The numerator $y^* S_p x$ is the signed projection of the physical action onto the mode. The denominator $y^* S_s x$ is the dynamic stiffness of the reduced eigenproblem at that pole; it becomes small or

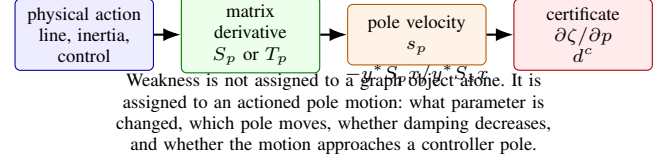


Fig. 5. Local certificate pipeline. The graph enters through S_p or T_p , but the decision is made after the pole and damping-ratio derivatives are computed.

strongly controller-shaped near a PLL or inverter-control pole. Equation (43) then converts pole motion into the operational quantity used by protection and planning engineers: damping ratio. In short,

$$\text{action } p \longrightarrow S_p \longrightarrow \frac{\partial s_k}{\partial p} \longrightarrow \frac{\partial \zeta_k}{\partial p} \quad (44)$$

is the complete chain from an equipment decision to a damping-margin decision. If a screening method skips the middle two arrows, it has not diagnosed a weak link; it has only ranked a static object.

The deliberately provocative engineering message is therefore narrow but sharp: “more network” is not the same statement as “more damping,” and “high modal visibility” is not the same statement as “best actuator.” A reinforcement, inertia change, or controller retuning must be judged by the pole it moves and by the controller pole it moves toward or away from.

IV. WEAK ELEMENTS ARE DYNAMIC CERTIFICATES

Let p_e be an admissible strengthening action on a line e , and let p_i be an admissible local action at bus i . Examples include increasing line admittance, adding damping, changing virtual inertia, or retuning a controller. For each action, define the local damping-margin derivative

$$g_p(k) = \frac{\partial \zeta(s_k)}{\partial p} \quad (45)$$

for the pole family of interest. A positive g_p means the action improves the damping ratio of that tracked pole; a negative value means the action damages it.

To distinguish ordinary damping-ratio geometry from control-resonant weakness, define the controller-pole distance and self-energy fraction

$$d_k^c = \min_{\mu \in \text{spec}(A_{cc})} \frac{|s_k - \mu|}{|s_k| + |\mu| + \epsilon}, \quad (46)$$

$$\phi_k^c = \frac{|y_k^* A_{ec} (s_k I - A_{cc})^{-2} A_{ce} x_k|}{|y_k^* S_s(s_k) x_k| + \epsilon}. \quad (47)$$

Here d_k^c measures proximity to condensed-control poles, while ϕ_k^c measures how much of the NEP denominator is supplied by the frequency-dependent control term.

Definition 1 (Controller-aware weak link). *A line e is weak for pole s_k and action direction p_e if*

$$g_{p_e}(k) < 0. \quad (48)$$

It is control-resonant weak if, additionally, d_k^c decreases under the action and ϕ_k^c or the self-energy contribution dominates the retained-network terms in (35).

Definition 2 (Controller-aware weak node). *A bus i is weak for a pole family if at least one physically admissible local action p_i gives $g_{p_i}(k) < 0$, or if the bus strongly participates in a mode with high ϕ_k^c and small d_k^c . Thus a node is not weak in isolation; it is weak relative to a mode, an action, and a controller configuration.*

These definitions intentionally do not use graph centrality. The following proposition states why.

Proposition 1 (Graph centrality cannot identify controller-aware weak elements). *Consider two linearized IBR grids with the same retained network graph and the same line centralities. Suppose $A_{ec} A_{ce} \neq 0$ for a retained mode s_k . Then, by changing only the condensed controller block A_{cc} and its residues while holding the graph fixed, the sign and magnitude of $\partial \zeta(s_k) / \partial p_e$ in (43) can change. Therefore no score that is a function only of the static graph can determine controller-aware weak links or weak nodes in general.*

Proof. For fixed graph and fixed retained action derivative, expand the Schur self-energy around the condensed-control poles:

$$A_{ec}(sI - A_{cc})^{-1} A_{ce} = \sum_j \frac{R_j}{s - \mu_j}, \quad (49)$$

where $\mu_j \in \text{spec}(A_{cc})$ and R_j are residues determined by controller couplings. The pole sensitivity (35) contains both the self-energy numerator and the denominator derivative

$$A_{ec}(sI - A_{cc})^{-2} A_{ce} = \sum_j \frac{R_j}{(s - \mu_j)^2}. \quad (50)$$

Moving a controller pole μ_j near s_k , or changing the phase/sign of the associated residue through controller parameters, changes the complex direction of s_p . Through (43), this can make $g_{p_e}(k)$ positive or negative while the graph, line centralities, and retained Laplacian are unchanged. Hence graph-only centrality cannot determine the weak-element sign. $\square$

TABLE I
WEAK-ELEMENT OBJECTS AND WHAT THEY CAN CERTIFY.

Object	Depends on	Missing for IBR damping
Graph centrality	Static topology	Controller poles and action direction
Line loading	Operating flow	Pole movement and damping ratio
Modal participation	Current eigenvector	Directional action derivative
Frequency-only score	line $\partial \text{Im}(s) / \partial p$	Decay-rate and self-energy terms
NEP weak certificate	$S_p, S_s, \text{spec}(A_{cc})$	Local; needs validation for finite actions

Remark 1. *The proposition does not say that centrality is useless. It says that centrality is incomplete for IBR damping margins when controller poles and network modes interact. A central line may be harmless, and a non-central line may be dangerous, if the latter moves a retained mode toward a controller resonance.*

A. Why Participation Is Not Enough

Modal participation is closer to dynamics than graph centrality, but it is still not the complete weak-element object. A participation factor answers “where does the current mode live?” The planning question is different: “what happens to the pole if I change this element?” The second question requires the directional derivative (35). For a line $e = (i, j)$, a static modal score may be based on $(q_i - q_j)^2$ or on power-flow loading. Such scores do not include A_{cc} , so they cannot see that the same line action can move a mode toward or away from a PLL pole. For a bus i , a large component of a critical eigenvector does not distinguish between adding damping, changing virtual inertia, retuning a PLL, or reinforcing adjacent branches. Those actions correspond to different $S_p(s)$ and may move the same pole in different directions.

Table I summarizes the distinction. The proposed certificate does not replace every score in the table; it identifies when a score is missing the dynamic mechanism needed for a damping-margin claim.

V. CONTROL-RESONANT BRAESS MECHANISM

Classical Braess effects concern network performance degradation after adding capacity. For damping margins, there are two different mechanisms that should not be conflated.

The first is geometric. Because $\zeta = -\text{Re}(s)/|s|$, a line reinforcement can lower ζ if it raises oscillation frequency more than it improves decay. This can happen in scalar models and does not require controller resonance.

The second is controller-mediated. Let p_e be line strengthening, and decompose the numerator of (35) into

$$y^* S_{p_e}(s) x = A_e + B_e + C_e, \quad (51)$$

where A_e is the retained stiffness/frequency contribution, B_e is the retained-network residual or modal-rotation contribution,

and C_e is the condensed-control self-energy contribution. A strict control-resonant Braess certificate is

$$\frac{\partial \zeta(s_k)}{\partial p_e} < 0, \quad (52)$$

$$\frac{\partial d_k^c}{\partial p_e} < 0, \quad (53)$$

$$|C_e| > \gamma \max(|A_e|, |B_e|), \quad C_e \text{ has damaging sign}, \quad (54)$$

with the nearest condensed-control pole dominated by inverter/control states. In words, strengthening the line damages damping, pulls the network-family mode closer to a controller pole, and the self-energy term dominates the sensitivity. This is the Braess-like mechanism that is specific to controller-aware IBR dynamics.

Proposition 2 (Control-resonant Braess certificate). *For a smooth tracked network-family pole $s_k(p_e)$, conditions (52)–(54) certify a controller-mediated reinforcement reversal. The reversal is not explained by static stiffness alone: if $|C_e| > \gamma \max(|A_e|, |B_e|)$ with $\gamma > 1$, then the damaging first-order pole motion is dominated by the condensed-control self-energy.*

Proof. Smoothness gives the first-order expansion

$$\zeta(s_k(p_e + \Delta p)) = \zeta(s_k(p_e)) + \frac{\partial \zeta}{\partial p_e} \Delta p + O(\Delta p^2).$$

Condition (52) makes the first-order change negative for small strengthening $\Delta p > 0$. Condition (53) states that the same action decreases the distance to the closest condensed-control pole, so the resolvent terms in (49)–(50) grow rather than recede. Finally, (54) rules out a retained-network explanation at first order, because the self-energy contribution is larger than both retained terms by factor γ and has the damaging sign. Thus the observed local reversal is controller-resonant in the Schur-NEP sense. $\square$

A. Reproducible Audit Procedure

The certificate is operationalized as follows. First, compute the full linearization and partition it as in (29). Second, extract zeros of (30) by contour integration and match them to the full eigensolve only after extraction. Third, classify the pole family using the reconstructed condensed state

$$x_c(s_k) = (s_k I - A_{cc})^{-1} A_{ce} x_e(s_k),$$

and the ratio

$$\pi_c(s_k) = \frac{\|x_c(s_k)\|}{\|x_e(s_k)\| + \|x_c(s_k)\|}.$$

Fourth, perturb a physical line parameter, rerun the power flow and eigensolve, and compare the finite-difference pole motion with (35). Fifth, label a reversal resonant only if it satisfies (52)–(54) and the nearest condensed pole is controller dominated. This prevents ordinary frequency–damping tradeoffs from being mislabeled as a new Braess mechanism.

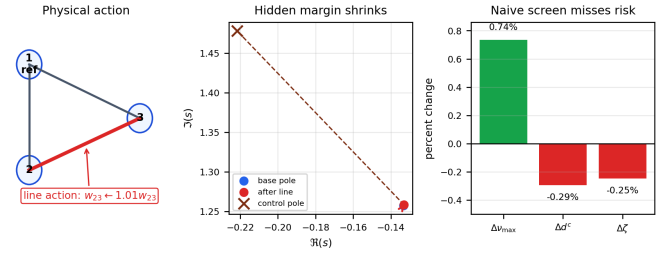


Fig. 6. Minimal three-bus witness. Strengthening line 2–3 increases the largest network stiffness, so a static synchronizing-torque screen sees a stronger graph. The controller-aware pole, however, moves toward a control pole and its damping ratio decreases.

VI. MINIMAL SYNTHETIC WITNESS

Before moving to IEEE 39-bus, it is useful to see the mechanism in a system small enough to inspect by hand. Consider the three-bus network in Fig. 6, with bus 1 used as the angle reference. The retained angles are $\theta = [\theta_2, \theta_3]^T$, and the grounded Laplacian is

$$L(w) = \begin{bmatrix} w_{12} + w_{23} & -w_{23} \\ -w_{23} & w_{13} + w_{23} \end{bmatrix}. \quad (55)$$

The retained network has $M = I$ and $D_0 = d_0 I$. A two-state controller is attached through

$$\ddot{\theta} + d_0 \dot{\theta} + L(w)\theta + Kz = 0, \quad (56)$$

$$\dot{z} = A_c z + B\dot{\theta}, \quad (57)$$

so that eliminating z gives

$$T(s) = s^2 I + s [d_0 I + K(sI - A_c)^{-1} B] + L(w). \quad (58)$$

The numerical witness uses

$$w_{12} = 0.8701, \quad w_{13} = 3.6556, \quad w_{23} = 3.9125, \\ d_0 = 0.7886, \quad \text{spec}(A_c) = -0.2216 \pm j1.4783.$$

The line action is a one-percent strengthening of the coupling line, $w_{23} \leftarrow 1.01w_{23}$.

A static synchronizing-stiffness screen sees only

$$\nu_{\max}(L) : 10.3284 \rightarrow 10.4044,$$

an increase of 0.74%. In that graph-only view, the network is simply stronger. The full controller-aware pole tells the missing part:

$$s : -0.133247 + j1.257268 \rightarrow -0.133005 + j1.258115,$$

so

$$\zeta : 0.105391 \rightarrow 0.105132, \quad \Delta\zeta = -2.59 \times 10^{-4}.$$

At the same time, the distance from the tracked pole to the nearest controller pole falls from 0.23804 to 0.23734. Thus the perturbation strengthens the graph but moves the dynamic pole toward the controller resonance and reduces damping.

This example is not presented as a realistic benchmark. It is a microscope for the mechanism. It shows why the paper does not define weak links by edge centrality or by stiffness increase alone. The relevant quantity is the pole motion of the coupled network–controller object.

TABLE II
CONTROLLER-AWARE SCHUR-NEP BRIDGE VALIDATION AGAINST FULL ANDES EIGENSOLVE.

Case	Critical pole	Family	π_c	Error
no-PSS	$-1.34635 + j8.61104$	I	0.680	$4.9 \times 10^{-14}\%$ f
base	$-1.34601 + j8.61148$	I	0.653	$1.6 \times 10^{-13}\%$ f
Mix60	$-10.7514 + j86.3419$	II	0.853	$8.5 \times 10^{-11}\%$ f

The damping-ratio errors are $6.3 \times 10^{-13}\%$, $1.1 \times 10^{-12}\%$, and $2.5 \times 10^{-9}\%$, respectively. Mix60 is a 13.742 Hz PLL-family mode; its nearest condensed-control pole is $-10.5593 + j86.246$ at distance 0.215.

TABLE III
CONTROL-RESONANT LINE-REINFORCEMENT AUDIT.

Metric	Value
Line/mode perturbations completed	460
Global-margin reversals	348
Network-family mode reversals	232
Strict resonant records	40
Median analytic-vs-NEP relative error	8.08×10^{-4}
Clearest case	Mix60/no-PSS, Line 20–34
Margin ζ	$0.123566 \rightarrow 0.123558$
Network-mode ζ	$0.133884 \rightarrow 0.133816$
Distance to matched A_{ce} pole	$3.33227 \rightarrow 3.32790$
A_e term	-1.09×10^{-9}
B_e term	7.97×10^{-9}
C_e term	-6.87×10^{-3}
Control denominator fraction	0.959

VII. IEEE 39-BUS ANDES EVIDENCE

The numerical evidence uses versioned ANDES IEEE 39-bus artifacts. The full ANDES eigensolve is the reference. The Schur NEP is solved by Beyn contour integration, not by fixed-point iteration. Zeros are extracted from contour moments and then locally polished by solving the intrinsic condition that the smallest singular value of $S_e(s)$ vanish; no ANDES target pole is supplied to the solver.

Table II gives the bridge gate. In no-PSS and base cases, the critical poles are network-family modes. In Mix60, the damping-margin pole is a high-frequency control-family mode dominated by PLL states. This is the failure mode of scalar graph reductions: the critical object is not contained in a static Laplacian or nodal damping coefficient.

After the bridge passed, each tested line reinforcement reduced the physical ANDES branch r and x by $1/(1+\epsilon)$ with $\epsilon = 0.01$, reran power flow, the full eigensolve, and the Schur-NEP contour solver. The audit covered Mix60, Mix60/no-PSS, and a soft-GFL negative-control profile. Table III summarizes the result.

The clearest case satisfies the strict certificate (52)–(54). The damaging sensitivity is not a frequency-only effect: the retained stiffness term is negligible, while the condensed-control self-energy term sets the sign and magnitude. The finite-difference network-mode derivative is -6.860×10^{-3} , and the analytic decomposition gives -6.874×10^{-3} , a relative error of 2.0×10^{-3} . The nearest condensed pole is PLL-dominated, with leading states from PLL1 and inverter-control channels.

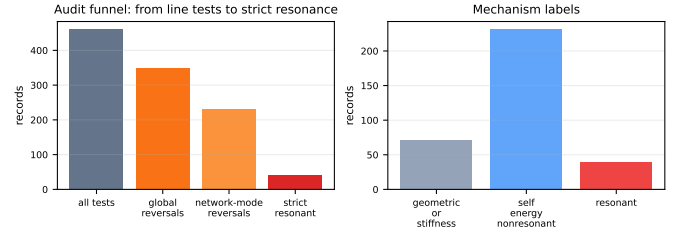


Fig. 7. Audit-level application view. Many line reinforcements reduce damping, but only a strict subset satisfies the controller-resonance certificate. This avoids confusing ordinary damping-ratio geometry with the proposed IBR-specific mechanism.

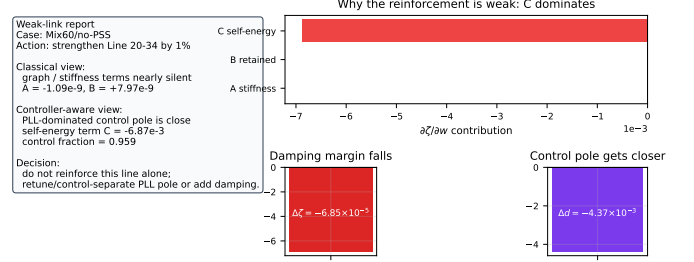


Fig. 8. Engineering report for the clearest weak link. A stiffness-only or graph-only view is nearly silent, but the controller-aware decomposition shows that the self-energy term dominates, the damping margin falls, and the mode moves closer to a PLL-dominated control pole.

The negative-control profile matters. Under the soft-GFL profile, the same spectral condition is removed: the network/control pole distance is much larger and the control-frequency denominator fraction falls. The audit therefore supports a mechanism claim, not a universal paradox claim. The resonant cases are modal-local and control-conditional; in these artifacts, the global margin is often still set by a control-family pole.

This is the practical failure mode the paper is meant to expose. A planner who sees only topology, loading, or retained stiffness would not flag Line 20–34 as a hidden damping-risk action in this operating point: the retained terms A_e and B_e are almost zero. The controller-aware certificate instead reports that the line action reduces damping because it decreases the separation between the network mode and a PLL-dominated pole. The recommended robustness action is therefore not “add more copper on that branch” by itself, but to retune or separate the implicated control pole, add damping, or choose a line action whose NEP sensitivity improves ζ without reducing d_k^c .

A. What This Shows Against Static Rankings

The ANDES evidence directly targets the title claim. The Mix60/no-PSS clearest case is not identified as weak because it is globally central in a graph-theoretic sense. It is identified as weak because the physical line-strengthening action has a negative NEP damping derivative and reduces the distance from a network-family pole to a PLL-dominated condensed-control pole. A graph-only score has no variable corresponding

TABLE IV
IEEE 39-BUS INERTIA WITNESS: THE MOST PARTICIPATING MACHINE IS NOT THE BEST INERTIA ACTION.

Action	Bus	M change	Post $\zeta_{\min}$	$\Delta\zeta$
Most participating GENROU_1	30	8.400 $\rightarrow$ 8.484	0.153794	-6.36×10^{-4}
Best audited GENROU_8	37	4.860 $\rightarrow$ 4.9086	0.154848	$+4.18 \times 10^{-4}$

to the distance d_k^c or the self-energy fraction ϕ_k^c ; therefore it cannot distinguish this line from a line with identical static graph role but different controller tuning.

The same logic applies to weak nodes. In an IBR grid, a bus may be a weak node for one intervention and not for another. If a local damping action moves the pole left, the bus is a useful damping-placement candidate. If a virtual-inertia or topology action rotates the pole toward a controller resonance, the same bus can be harmful for that action. The certificate therefore changes weak-node language from a scalar ranking to a typed statement:

bus i is weak for mode k under action p_i .

This typed form is less convenient than centrality, but it is the form that remains meaningful when controller dynamics set the margin.

B. Inertia Witness: Participation Can Pick the Wrong Machine

The same warning appears for inertia placement. In the IEEE 39-bus base case, the critical network-family pole has its largest state participation at GENROU_1: $\omega_{\text{GENROU 1}}$ contributes 0.3682 and $\Delta_{\text{GENROU 1}}$ contributes 0.3422. A common modal-participation heuristic would therefore add inertia at that machine, located at bus 30.

The controller-aware NEP and the full ANDES eigensolve give the opposite decision. Increasing the documented inertia parameter GENROU.M by one percent at bus 30 reduces the damping margin. Increasing inertia at GENROU_8, bus 37, improves it. Table IV summarizes the witness.

The local NEP derivative predicts the damaging sign at bus 30: $\partial\zeta/\partial\epsilon = -6.47 \times 10^{-2}$, while the full ANDES finite difference gives -6.36×10^{-2} . This is the cleanest weak-node message: “where the mode lives” and “where an action improves damping” are different questions. The result should not be overextended. It is a documented GENROU.M perturbation in full ANDES, not a claim about adding a new generator with its own dispatch and controls, and not a validated GFM virtual-inertia result.

VIII. RATIONAL-FILTER INTERPRETATION

The same conclusion can be stated in graph-filter language. A polynomial graph filter $h(L)$ can reweight Laplacian modes, but it cannot create poles that are not in the graph operator. The Schur self-energy is different. In modal coordinates of the retained network, its entries are

$$\hat{\Sigma}_{ab}(s) = q_a^* A_{ec} (sI - A_{cc})^{-1} A_{ce} q_b, \quad (59)$$

TABLE V
NOVELTY BOUNDARY FOR THE PROPOSED CONFERENCE PAPER.

Known ingredient	New use in this paper
Schur complement and NEP sensitivity	Define weak elements as action-conditioned pole certificates
Graph/modal participation	Shown insufficient without controller-pole distance
Braess-style reversal language	Separated into geometric vs. control-resonant damping mechanisms
Beyn contour integration	Used as non-circular bridge validating the reduced object against ANDES

so the filter depends on the evaluation point s and on the condensed-control spectrum. Using the residue form (49),

$$\hat{\Sigma}_{ab}(s) = \sum_j \frac{q_a^* R_j q_b}{s - \mu_j}. \quad (60)$$

This is a rational filter over the network-controller interconnection, not a polynomial filter of the graph alone. It has two consequences.

First, weakness is mode-pair dependent. A line action may be benign for one network mode and harmful for another because the residues $q_a^* R_j q_b$ differ by mode pair. Second, weakness is frequency dependent. The same graph and the same line action can become dangerous only when the tracked pole lies near a controller pole. This explains why a static ranking can look reasonable in a synchronous-machine regime and fail in an IBR regime without any topological change.

The conference-level innovation is therefore not a new graph signal-processing algorithm. It is the identification of the correct signal path for damping-margin weakness:

$$\text{action} \rightarrow S_p(s) \rightarrow \Delta s_k \rightarrow \Delta \zeta_k, \quad (61)$$

$(sI - A_{cc})^{-1}$ retained.

Dropping the final resolvent collapses the problem back to graph centrality or modal participation and removes the mechanism observed in Table III.

IX. DISCUSSION

The proposed definitions shift weak-element analysis from graph ranking to mechanism attribution. A graph centrality score can say that a bus or line is structurally important, but it cannot say whether strengthening it increases damping, decreases damping by geometry, or decreases damping by pulling a network-family pole toward a PLL resonance. Those are different engineering problems. The first can be addressed by conventional reinforcement. The second requires damping-aware line ranking. The third may require controller retuning, pole separation, or avoiding a reinforcement that is attractive under static metrics.

A. What This Gives to Power Engineers

The practical output is a diagnostic workflow, not a decorative graph metric. For each proposed action, the engineer obtains four quantities:

- 1) the tracked pole s_k and its family, network-dominated or controller-dominated;

- 2) the damping derivative $g_p = \partial\zeta(s_k)/\partial p$;
- 3) the controller-pole distance d_k^c and whether the action shrinks it;
- 4) the dominant term in the sensitivity numerator, retained-network or self-energy.

Those four numbers answer a planning question that centrality cannot answer: should the action be accepted, rejected, or replaced by a controller retuning? In the Line 20–34 witness, the answer is not merely “this line is weak.” The answer is: this line is weak for the tracked network-family mode under a strengthening action because the action reduces damping and moves the mode toward a PLL-dominated control pole. That statement is directly actionable: avoid that reinforcement in that operating point, separate the implicated pole, or choose an alternative line/control action whose certificate has the opposite sign.

The controversial conclusions are deliberately stated with their scope:

- A stronger grid can be a weaker damping design if the added stiffness pulls a mode toward a controller pole.
- The most visible machine in a mode is not necessarily the best place to add inertia; visibility is not a signed action derivative.
- A stable IBR operating point can carry hidden margin debt: the pole is stable, but already too close to controller dynamics for a static screen to be trusted.

These are not universal laws. They are local, model-checked warnings. That is precisely why the paper uses full eigensolve and Schur-NEP certificates rather than a new centrality score.

The paper also clarifies what is not claimed. First, the NEP bridge is validated in three diagnostic IEEE 39-bus cases, not across all possible IBR dispatches. Second, the resonant Braess mechanism appears in a minority of strict line/mode records; it is not the dominant cause of every damping decrease. Third, the result is not a replacement for full ANDES eigensolve. It is a reduced diagnostic object that preserves the controller pole responsible for the mechanism.

For this conference-method contribution, this is the intended novelty boundary: established tools such as Schur complements, contour NEP solvers, eigenvalue sensitivities, and damping ratios are assembled into a controller-aware definition of weak nodes and weak links. The new power-systems statement is that IBR weak elements are action- and mode-conditioned dynamic certificates, not graph-centrality objects.

X. CONCLUSION

Weak nodes and weak links in IBR grids should not be defined solely by static graph centrality. After controller states are condensed, the damping-margin problem is governed by a rational Schur NEP whose self-energy contains PLL and inverter-control poles. A local line or node action is weak when it moves a relevant pole in a damaging damping direction, especially when it also reduces the distance to a condensed-control pole. This framework explains a control-resonant Braess-like mechanism: line strengthening can reduce

damping because it pulls a network-family mode toward a PLL-dominated controller pole. IEEE 39-bus ANDES artifacts validate the bridge and observe this mechanism under strict criteria.

The practical conclusion is uncomfortable but useful: a planning action that looks robust in a graph can be a damping-risk action in the dynamics. More copper, more inertia, or higher modal participation are not certificates of stability. The certificate is the signed motion of the damping-margin pole. The next step is not to invent another centrality score, but to benchmark these certificates against conventional weak-node and weak-link rankings over broader calibrated IBR operating ranges.

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